Musa Azeem

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Education

University of South Carolina Honors College — Columbia, SC
Bachelor of Science in Computer Engineering; Overall GPA: 4.000 — May 2024
Summa Cum Laude; President's Honor List, all semesters

University of South Carolina — Columbia, SC
Masters of Science in Computer Science; Overall GPA: 4.000 — May 2025

Experience

Cognito Forms — Columbia, SC
Full-stack Software Engineer — Jun 2025 – Present

- Develop and maintain the Cognito Forms web app across a .NET Core backend and TypeScript/JavaScript + Vue.js frontend.
- Deliver bug fixes and performance improvements, optimizing algorithms to reduce latency and memory usage.
- Develop innovative features, including integrated LLM-based and payment services using Stripe, PayPal, and Square SDKs.
- Run delivery workflows with Azure DevOps pipelines, automated agent tasks, and production diagnostics in Application Insights.

University of South Carolina — Columbia, SC
Artificial Intelligence Research Assistant — Dec 2021 – May 2025

- Led the research and implementation of deep learning models using PyTorch (LSTM, CNN, Autoencoders), achieving high accuracy in human activity recognition, rodent sleep status prediction, and protein structure prediction, resulting in multiple publications.
- Engineered an ETL pipeline to process EEG data from SQL databases for analysis and machine learning; deployed on a server for real-time processing.
- Developed a Python library and ML models to evaluate protein structure predictions without experimental structures, producing an accuracy metric comparable to RMSD.
- Built and deployed full-stack solutions including an Android WearOS app for real-time accelerometer streaming into an edge ANN and a Flask web app for prediction workflows.

SIOS Technology Corporation — Columbia, SC
AI Solutions Architect & Software Engineering Intern — May 2023 – Feb 2024

- Led end-to-end development of a RAG-powered AI chatbot to deliver accurate, summarized answers based on information pulled from documentation, Salesforce tickets, and internal notes. Developed with a quantized local Llama-3 instance hosted via llama.cpp, an auto-updated vector database, and website/documentation parsing.
- Created a Python library to parse customer logs, extract critical information, detect error patterns, and present results through an interactive web interface.

Capgemini — Houston, TX
Rise Intern - Technology Consultant Role — Aug 2023

- Directed a rapid turnaround case study to deliver a mobile app launch strategy, functional prototype, and business model.
- Engaged with industry experts to gain insights into software engineering and global technology consulting processes.

Skills

Programming Languages: Python, C#, C/C++, TypeScript, JavaScript, R, Kotlin, SystemVerilog
Tools/Libraries: PyTorch, TensorFlow, .NET Framework, .Net Core, Pandas, NumPy, Flask, Matplotlib, Biopython, React, Vue.js, Express, Android SDK
Machine Learning: Reinforcement Learning, Random Forest, Clustering Algorithms, Model Evaluation & Diagnostics
Deep Learning: CNN, LSTM, Residual Networks, Autoencoder Architecture, Model Regularization
Data Analysis: SQL, Data Processing & Visualization, Exploratory Data Analysis, ETL Pipelining
Other: Claude Code, Generative AI, Git, Linux, AWS, Docker, Embedded Systems, Microcontroller, Circuit Design, Chip Design

Publications

[1] Musa Azeem, Christopher Lee, Aaron Hein, C. Christopher Ott, and Homayoun Valafar, "Reformulation of the protein databank for real-time search of geometrical attributes of protein structures," Frontiers in Molecular Biosciences 12: 1694750. https://doi.org/10.3389/fmolb.2025.1694750

[2] Andrew Smith, Musa Azeem, et al., "Toward Concurrent Identification of Human Activities with a Single Unifying Neural Network Classification: First Step," Sensors 24, no. 14: 4542. https://doi.org/10.3390/s24144542

[3] Grant King, Musa Azeem, Savannah Noblitt, Ramtin Zand, and Homayoun Valafar, "Rep Smarter, Not Harder: AI Hypertrophy Coaching with Wearable Sensors and Edge Neural Networks." https://doi.org/10.1109/ICMLA66185.2025.00179

[4] Musa Azeem and Homayoun Valafar, "Dihedral Angle Adherence: Evaluating Protein Structure Predictions in the Absence of Experimental Data," in 2024 Congress in Computer Science, Computer Engineering, & Applied Computing (CSCE), Las Vegas, NV, USA, 2024. https://doi.org/10.1007/978-3-031-85902-1_28

Awards & Engagement

NASA STEM Outreach Award Aug 2023 – May 2024
Secured funding to conduct research and lead eight outreach events, teaching high school students across South Carolina the basics of web programming. Developed a universal smartwatch application to record sensor data and facilitate event annotation.

Outstanding Senior in Computer Engineering Mar 2024
Recognized for major achievement in service, leadership, academics, and research.

Projects

Deep Learning Eating Detection (github.com/musa-azeem/eating-detection) Jul 2023 – May 2024
Researched ML techniques to use tri-axis accelerometer data to recognize eating gestures. Built PyTorch models including LSTM, CNN, Residual, and Autoencoder networks.

Protein Analysis (github.com/musa-azeem/pdbmine_analytics_lib) May 2024 – May 2025
Developed a Python library and advanced data analysis and ML techniques to evaluate protein structure predictions in the absence of experimentally determined structures. Used to calculate an accuracy metric comparable to the industry standard RMSD, providing robust evaluation capabilities.

Faculty Dashboard Full Stack App (github.com/SCCapstone/K3MS) Aug 2023 – May 2024
Built a full-stack dashboard for UofSC faculty using Flask, React, and MySQL. Implemented REST APIs, CSV ingestion, data processing, and analysis; deployed with Apache.

AI Fitness Coaching (github.com/musa-azeem/RepAI) Jan 2025 – present
Developed a ResNet and LSTM based model to identify near-failure states in real time during resistance training using a single wrist-mounted IMU.